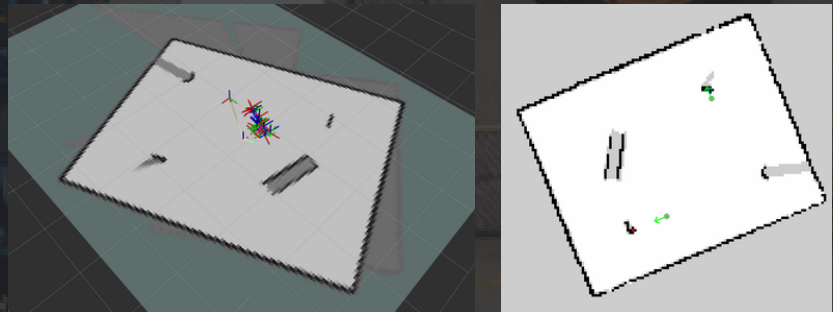


## Design choices & Algorithm

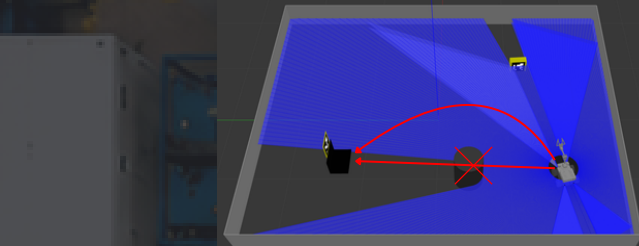
### Localization & Mapping

The surroundings are mapped using Mirte's built-in Lidar with the SLAM algorithm. Moreover, using computer vision, the distance and yaw of the stations can be captured by the front camera, where the relative location of the stations to the robot and the SLAM map can be computed, stored, and used for navigation

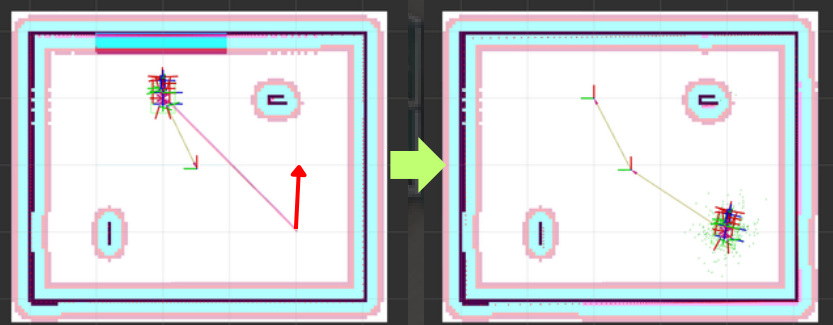


### Navigation & Planning

Taking the SLAM map and the recorded stations A and B coordinates as input, the robot will use the SLAM map to create the global costmap. Taking the coordinates of A and B, using the nav2 algorithm, the robot travels to the designated coordinates, minimizing the travel distance and possibilities of obstacle collision.



While directing the robot, local costmaps are used to provide real-time lidar updates for the robot, in case a new obstacle appears to the robot that wasn't previously captured by the SLAM map. It is also worth highlighting that before



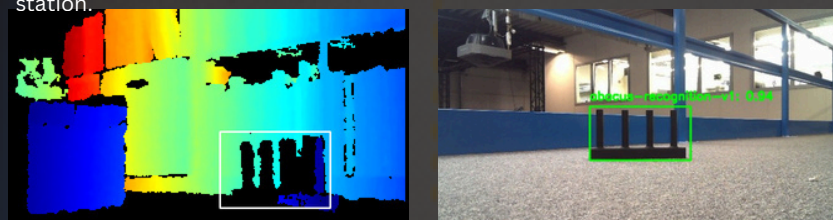
### Detection, Tracking & Segmentation

Two computer vision approaches are taken for station detection:

Station A is detected and located using QR code recognition with the Aruco package; the distance and yaw of the station depend on the orientation of the QR code.



Station B is detected and located using the YOLO CNN algorithm, where a dataset of 70 images is used to train the model. The robot is able to detect the silhouette of the abacus in combination with the depth camera to locate the station.



For the clock recognition and reading, the OCR package (YOLO) is implemented.

# Cognitive Robotics: Group 5

## Problem Statement

“How can a mobile robot autonomously **navigate** to a visual information source, **interpret** the displayed information, react, process, and **transfer** that information into a **physical representation** at a second location?”

## Problem Setup

The task involves a robot navigating a structured environment where it must extract meaningful information from one location and translate it into a physical representation in another. Specifically, the robot will observe a visual input (such as a clock display), interpret the information it perceives, and encode that information by placing objects in certain positions.

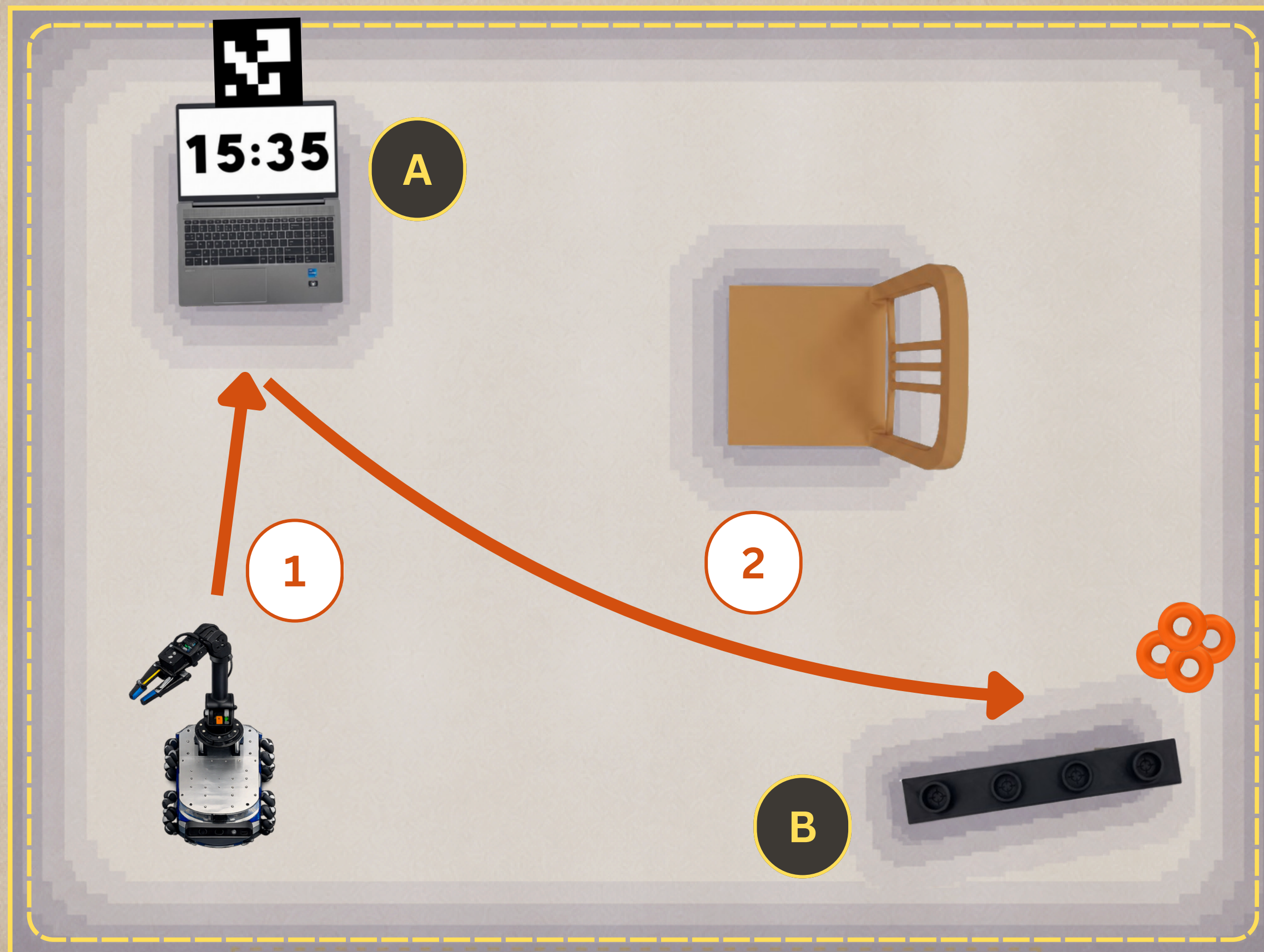
Station A consists of a laptop displaying a time; the robot has to detect and read the clock using computer vision and store the value.

### Station A



Station B consists of an abacus. Using the time data from station A, the robot has to place the corresponding rings on the abacus.

### Station B



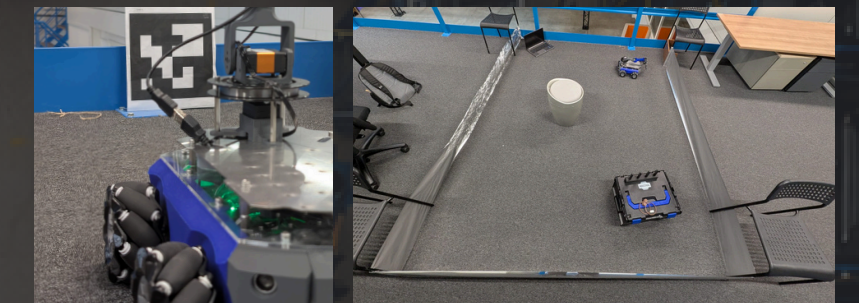
## Mission design & Approach

### General Mission

The mission of the robot is separated into two stages: the preparation stage and the demo stage. The former prepares the robot to acquire the SLAM map and station locations, and the latter operates the robot to navigate and achieve the intended tasks.

### Trial Stage

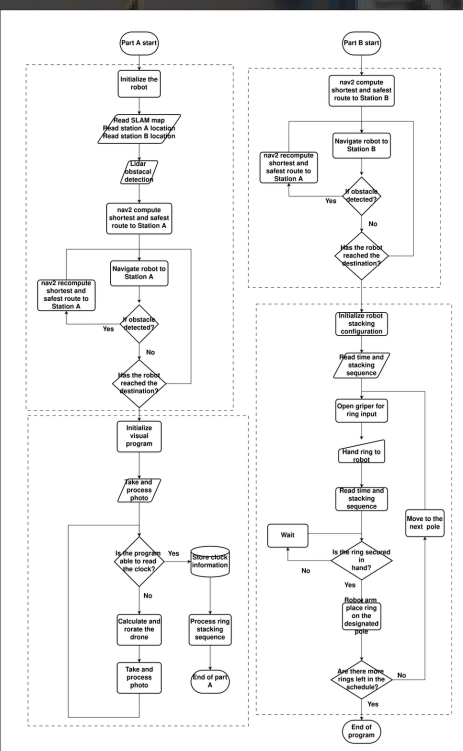
In this stage, the robot is manually traversed around the world to perform SLAM mapping and station recognition. Stations A and B are detected using computer vision, where their relative positions to the SLAM map and the robot's destination point to reach the stations are computed and stored with the completed SLAM map.



### Demo Stage

In this stage, the robot performs the tasks using the acquired SLAM map and station coordinates. The following steps are taken:

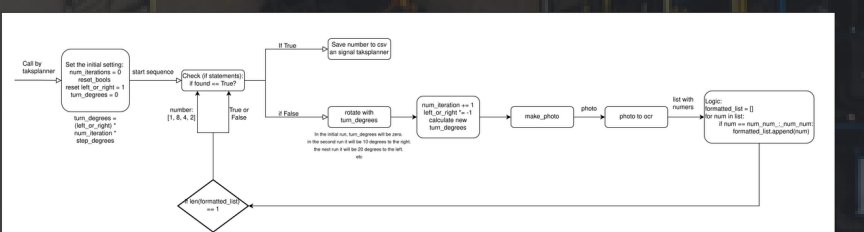
1. Initiate robot, calibrate lidar readings, navigate to station A according to the SLAM map and pre-recorded coordinates, and read the displayed clock info.
2. When clock reading is accomplished, the robot then navigates to station B, after aligning with the station, the robot will pick up rings to place onto the abacus based on the previous clock reading.



The flow diagram on the left depicts the general workflow of the robot during this stage. At the start, the robot reads the SLAM map and station locations. The mission is separated into two parts: part A for clock recognition and processing, part B for abacus ring stacking.

Using the locations, the robot would navigate to station A first, to read and store the clock information (see specific flowchart below).

The robot will then navigate to station B after the clock reading, and it will initiate the stacking sequence based on the clock information previously stored. The robot gripper will be actuated to stack the rings at the designated locations.



## Performance & Evaluation

Station recognition accuracy quantifies the robot's performance. Using a dataset of 10 tests, the accuracy parameters of the Aruco and YOLO (abacus) algorithms used for the QR code and abacus recognition are presented in the tables below

| Method        | Mean absolut e error | Max absolut e error | Mean % error | Max % error |
|---------------|----------------------|---------------------|--------------|-------------|
| YOLO (abacus) | 3.27 cm              | 3.80 cm             | 3.24%        | 6.96%       |
| ArUco         | 1.09 cm              | 1.50 cm             | 0.99%        | 1.59%       |

All in all, both methods are able to consistently predict the distance from the camera to the stations. Aruco is more robust since QR codes with white backgrounds are easier to identify. The YOLO method has a higher overall error, but this could be improved by implementing more training sets.

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